
Abstract

Keywords:

1. Introduction

2. Regime-Local Reduced-Order Representations

The Regime-Adaptive Localized Mixture-of-Experts Reduced-Order Model (RAL-MoE-ROM) represents a parametrized flow by a collection of local reduced dynamical systems rather than by a single global coordinate system. The construction is motivated by the fact that the steady, Hopf-onset, and periodic regimes occupy geometrically and dynamically distinct parts of the solution manifold. A local chart therefore contains not only a pair of POD bases, but also its own mean fields, coefficient scalars, physical operators, history convention, time integrator, and algebraic pressure closure. This section defines these local representations and the interfaces that may be used between adjacent charts.

2.1. Parameterized flow dynamics and regime decomposition

Let $\Omega \subset \mathbb{R}^2$ denote the spatial domain and let $\mu = \text{Re} \in \mathcal{P}$ be the Reynolds-number parameter. The high-fidelity state at time t is written as

$$\mathbf{x}(t; \mu) = (\mathbf{u}(t; \mu), p^\circ(t; \mu)), \quad (1)$$

where $\mathbf{u}$ is the velocity field and p° is a gauge-corrected pressure field. The parameter domain is covered by three regime-local sets,

$$\mathcal{P} \subseteq \mathcal{P}_S \cup \mathcal{P}_H \cup \mathcal{P}_P, \quad r \in \mathcal{R} := \{S, H, P\}, \quad (2)$$

associated with steady, Hopf-onset, and periodic dynamics. The cover is not required to be disjoint. Its adjacent overlaps are precisely the locations at which more than one local model may be considered:

$$\mathcal{O}_{SH} = \mathcal{P}_S \cap \mathcal{P}_H, \quad \mathcal{O}_{HP} = \mathcal{P}_H \cap \mathcal{P}_P. \quad (3)$$

The certified topology does not contain a direct S – P edge. Hence, physical proximity of two parameters alone does not authorize a cross-chart prediction; adjacency and dynamical admissibility are both required.

Formally, chart r is the contract

$$\mathcal{C}_r = (\bar{\mathbf{u}}^{(r)}, \Phi_u^{(r)}, \bar{p}^{(r),\circ}, \Phi_p^{(r)}, \mathcal{S}_r, \mathcal{G}_r, \mathcal{P}_r, \mathcal{I}_r), \quad (4)$$

where $\mathcal{S}_r$ collects the local scalars and feature contract, $\mathcal{G}_r$ is the local velocity Galerkin operator, $\mathcal{P}_r$ is the Pressure–Poisson asset, and $\mathcal{I}_r$ contains the native integrator together with its history and phase

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conventions. Equal modal dimensions across the three charts imply equal vector lengths, not a shared coordinate system.

For each regime r , complete Reynolds-number trajectories are assigned to training, validation, and test populations. All snapshots of one trajectory remain in the same population. The local reduced assets are constructed from the training trajectories of that regime only. This complete-trajectory partition prevents a history window or a later segment of a trajectory from appearing in a different population from its initialization.

2.2. Area-weighted local POD coordinates and pressure gauge

Let N_u and N_p denote the velocity and pressure degrees of freedom on the common finite-volume mesh. The discrete physical inner products are

$$\langle \mathbf{v}, \mathbf{w} \rangle_{M_u} = \mathbf{v}^\top M_u \mathbf{w}, \quad \langle q, s \rangle_{M_p} = q^\top M_p s, \quad (5)$$

where M_u and M_p contain the same cell-area weights used when the corresponding POD bases were constructed. In particular,

$$M_p = \text{diag}(w_1, \dots, w_{N_p}), \quad M_u = \text{diag}(M_p, M_p), \quad (6)$$

for the two velocity components. For chart r , the weighted POD bases

$$\Phi_u^{(r)} \in \mathbb{R}^{N_u \times d_u}, \quad \Phi_p^{(r)} \in \mathbb{R}^{N_p \times d_p}, \quad d_u = d_p = 32, \quad (7)$$

satisfy

$$\Phi_u^{(r)\top} M_u \Phi_u^{(r)} = I_{d_u}, \quad \Phi_p^{(r)\top} M_p \Phi_p^{(r)} = I_{d_p}. \quad (8)$$

Each chart also has independent training-set mean fields $\bar{\mathbf{u}}^{(r)}$ and $\bar{p}^{(r),\circ}$.

Pressure is made identifiable before projection. With cell-area vector $\mathbf{w}$, the gauge operator is

$$\mathcal{G}(p) = p - \frac{\mathbf{w}^\top p}{\mathbf{w}^\top \mathbf{1}} \mathbf{1}, \quad p^\circ := \mathcal{G}(p). \quad (9)$$

The same operator, area weights, and degree-of-freedom ordering are used in training, cross-chart projection, rollout reconstruction, and evaluation. Consequently, pressure coefficients are never compared or blended before the gauge convention has been enforced.

The raw physical modal coordinates in chart r are

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad (10)$$

$$\mathbf{b}^{(r)} = \Phi_p^{(r)\top} M_p (p^\circ - \bar{p}^{(r),\circ}). \quad (11)$$

The corresponding decoders are

$$\mathcal{D}_r^u(\mathbf{a}^{(r)}) = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (12)$$

$$\mathcal{D}_r^p(\mathbf{b}^{(r)}) = \bar{p}^{(r),\circ} + \Phi_p^{(r)} \mathbf{b}^{(r)}. \quad (13)$$

Coefficient standardization is chart-specific:

$$\widetilde{\mathbf{a}}^{(r)} = (\mathbf{a}^{(r)} - \boldsymbol{\mu}_a^{(r)}) \oslash \boldsymbol{\sigma}_a^{(r)}, \quad \widetilde{\mathbf{b}}^{(r)} = (\mathbf{b}^{(r)} - \boldsymbol{\mu}_b^{(r)}) \oslash \boldsymbol{\sigma}_b^{(r)}. \quad (14)$$

Here $\oslash$ denotes componentwise division. These normalized variables are internal neural-network coordinates. Physical projection, chart conversion, and field reconstruction always operate on inverse-scaled raw modal coefficients.

2.3. Mappings and interface contracts between adjacent charts

Analytic state maps are constructed only for adjacent ordered pairs

$$\mathcal{E}_{\text{adj}} = \{S \rightarrow H, H \rightarrow S, H \rightarrow P, P \rightarrow H\}. \quad (15)$$

For velocity, decoding a state in chart r and projecting it into chart k gives the affine map

$$\mathbf{a}^{(k)} = T_{k \leftarrow r}^u \mathbf{a}^{(r)} + \mathbf{c}_{k \leftarrow r}^u, \quad (16)$$

with

$$T_{k \leftarrow r}^u = \Phi_u^{(k)\top} M_u \Phi_u^{(r)}, \quad \mathbf{c}_{k \leftarrow r}^u = \Phi_u^{(k)\top} M_u (\bar{\mathbf{u}}^{(r)} - \bar{\mathbf{u}}^{(k)}). \quad (17)$$

The pressure map is analogously

$$\mathbf{b}^{(k)} = T_{k \leftarrow r}^p \mathbf{b}^{(r)} + \mathbf{c}_{k \leftarrow r}^p, \quad (18)$$

where

$$T_{k \leftarrow r}^p = \Phi_p^{(k)\top} M_p \Phi_p^{(r)}, \quad \mathbf{c}_{k \leftarrow r}^p = \Phi_p^{(k)\top} M_p (\bar{p}^{(r),\circ} - \bar{p}^{(k),\circ}). \quad (19)$$

The affine offsets in (17)–(19) account only for different chart means. They therefore apply to states, but not to increments or derivatives.

The maps are fixed offline operators, not trainable neural layers. No online matrix inverse is required. Their certification examines projection floors, physical-field and fluctuation errors, energy errors, round-trip errors, principal angles, and singular spectra. Small singular values identify directions that are private to one chart rather than shared by an overlap; this interpretation is more informative than a condition number alone. For example, the velocity principal angles satisfy

$$\cos \theta_i^u = \sigma_i(T_{k \leftarrow r}^u), \quad (20)$$

and the target-chart velocity projection floor is

$$e_{\text{floor}}^u(\mathbf{u}; k) = \frac{\|(I - \Phi_u^{(k)} \Phi_u^{(k)\top} M_u)(\mathbf{u} - \bar{\mathbf{u}}^{(k)})\|_{M_u}}{\|\mathbf{u} - \bar{\mathbf{u}}^{(k)}\|_{M_u} + \varepsilon}. \quad (21)$$

Round-trip error is evaluated by applying the two affine state maps in sequence and comparing the returned physical field with the original. These quantities diagnose representation compatibility; they are not trainable penalties in the frozen-specialist assembly.

A valid chart transfer acts on the complete physical history. With the three-state convention

$$\mathcal{H}_n^{(r)} = \{(\mathbf{a}_{n-\ell}^{(r)}, \mathbf{b}_{n-\ell}^{(r)}, t_{n-\ell})\}_{\ell=0}^2, \quad (22)$$

every state is inverse-scaled, mapped by (16) and (18), and then passed through the target specialist's native feature builder. The target Galerkin quantities, coefficient differences, phase variables when applicable, and pressure-closure history are reconstructed from the mapped states. A normalized feature tensor from the source specialist is never copied into the target specialist.

These interfaces do not define a global public POD space, do not authorize direct blending of modal coefficients, and do not create a common velocity–pressure vector field. Their role is limited to consistent state initialization and adjacent-chart diagnostics. The final hierarchical assembly in Section 4 preserves each local specialist's native dynamics and performs any certified soft combination only after decoding to physical fields.

3. Attractor-Specialized HPRS-MoE-ROM Specialists

Each regime chart is equipped with a complete local reduced-order model, referred to as a specialist. The three specialists share the Hybrid Physics-Residual Sparse Mixture-of-Experts Reduced-Order

Model (HPRS-MoE-ROM) design pattern, but retain regime-specific dynamical contracts. Within a specialist, the term expert refers only to a neural component of the sparse residual model; it does not denote the whole local ROM. This distinction separates the inner, channel-wise sparse mixture from the outer regime-level assembly introduced later.

3.1. Semi-explicit hybrid physics–data reduced dynamics

For regime r , the velocity coefficients satisfy the hybrid reduced dynamics

$$\dot{\mathbf{a}}^{(r)} = \mathbf{f}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) + \boldsymbol{\rho}_{\theta_r}^u(\boldsymbol{\xi}^{(r)}), \quad (23)$$

where $\mathbf{f}_r^G$ is the chart-local Galerkin velocity right-hand side and $\boldsymbol{\rho}_{\theta_r}^u$ is a learned residual. The fixed physical contribution has the polynomial form

$$\mathbf{f}_r^G(\mathbf{a}, \mathbf{b}; \mu) = \mathbf{c}_r(\mu) + A_r(\mu)\mathbf{a} + H_r : (\mathbf{a} \otimes \mathbf{a}) + P_r\mathbf{b}. \quad (24)$$

All tensors in (24) are constructed in the local basis of regime r . The residual is therefore a correction to a retained physical operator, rather than a replacement for it.

The velocity macro-step is expressed abstractly as

$$\mathbf{a}_{n+1}^{(r)} = \mathcal{I}_r(\mathbf{a}_n^{(r)}, \mathbf{f}_r^G + \boldsymbol{\rho}_{\theta_r}^u; \Delta t_r), \quad (25)$$

where $\mathcal{I}_r$ is the specialist’s native velocity integrator. The time step is supplied to the integrator; it is not a network input. The certified implementation uses explicit Euler for the steady specialist and classical fourth-order Runge–Kutta (RK4) for the Hopf and periodic specialists.

Pressure is not assigned a learned continuous-time derivative. After the velocity macro-step, a local Pressure–Poisson operator produces

$$\mathbf{b}_{n+1}^{PP,(r)} = \mathcal{P}_r(\mathbf{a}_{n+1}^{(r)}; \mu). \quad (26)$$

The steady specialist applies an additive algebraic correction,

$$\mathbf{b}_{n+1}^{(S)} = \mathbf{b}_{n+1}^{PP,(S)} + \boldsymbol{\rho}_{\theta_S}^p(\boldsymbol{\xi}_n^{(S)}), \quad (27)$$

whereas the Hopf and periodic specialists use an adaptive gate,

$$\mathbf{b}_{n+1}^{(r)} = \boldsymbol{\alpha}_n^{(r)} \odot \mathbf{b}_{n+1}^{PP,(r)} + \boldsymbol{\rho}_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)}), \quad \boldsymbol{\alpha}_n^{(r)} = \text{sigmoid}(\boldsymbol{\eta}_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)})), \quad r \in \{H, P\}. \quad (28)$$

Equations (23)–(28) define a semi-explicit velocity–pressure system: velocity is advanced dynamically, while pressure is recovered as an algebraic macro-step closure.

3.2. Sparse mixture-of-experts architecture and feature construction

The current-state feature vector combines reduced states, the fixed physical operator, the parameter, and chart-native descriptors. In schematic form,

$$\boldsymbol{\chi}_n^{(r)} = [\widetilde{\mathbf{a}}_n^{(r)}, \widetilde{\mathbf{b}}_n^{(r)}, \widetilde{\mathbf{g}}_n^{(r)}, \widetilde{\mu}, \widetilde{\mu}^{-1}, \boldsymbol{\varphi}_n^{(r)}, \mathbf{q}_n^{(r)}], \quad (29)$$

where $\mathbf{g}_n^{(r)} = \mathbf{f}_r^G(\mathbf{a}_n^{(r)}, \mathbf{b}_n^{(r)}; \mu)$, $\mathbf{q}_n^{(r)}$ contains chart-native physical descriptors, and $\boldsymbol{\varphi}_n^{(r)}$ is present only when required by the phase contract. For the steady and periodic specialists, the four indexed Fourier harmonics give the eight-dimensional encoding

$$\boldsymbol{\varphi}_n^{(r)} = [\sin(2\pi k \phi_{r,n}), \cos(2\pi k \phi_{r,n})]_{k=1}^4, \quad r \in \{S, P\}. \quad (30)$$

The phase $\phi_{r,n}$ is supplied by the registered native trajectory clock; it is not inferred from the predicted state. The Hopf specialist is phase-free, i.e., $\boldsymbol{\varphi}_n^{(H)} = \emptyset$.

The history length is three states: the current state and two previous states. For $\ell \in \{1, 2\}$, the history builder appends

$$\mathbf{h}_{n,\ell}^{(r)} = [\widetilde{\mathbf{a}}_{n-\ell}^{(r)}, \widetilde{\mathbf{b}}_{n-\ell}^{(r)}, \widetilde{\mathbf{g}}_{n-\ell}^{(r)}, \Delta\widetilde{\mathbf{a}}_{n,\ell}^{(r)}, \Delta\widetilde{\mathbf{b}}_{n,\ell}^{(r)}, \Delta\widetilde{\mathbf{g}}_{n,\ell}^{(r)}], \quad (31)$$

with

$$\Delta\widetilde{\mathbf{a}}_{n,\ell}^{(r)} = \widetilde{\mathbf{a}}_n^{(r)} - \widetilde{\mathbf{a}}_{n-\ell}^{(r)}, \quad (32)$$

and analogously for pressure and Galerkin features. These are raw coefficient differences; they are not divided by a time interval. The final network input is

$$\boldsymbol{\xi}_n^{(r)} = [\boldsymbol{\chi}_n^{(r)}, \mathbf{h}_{n,1}^{(r)}, \mathbf{h}_{n,2}^{(r)}]. \quad (33)$$

For the steady and periodic specialists, the current block has dimension 117 and each historical layer has dimension 192, hence

$$\dim \boldsymbol{\xi}^{(S)} = \dim \boldsymbol{\xi}^{(P)} = 117 + 2(192) = 501. \quad (34)$$

Removing the eight indexed-phase entries gives the Hopf common-backbone dimension 493; appending its 67 fluctuation and radial descriptors yields

$$\dim \boldsymbol{\xi}^{(H)} = 493 + 67 = 560. \quad (35)$$

The residual network uses two levels of sparsity. A group router first selects one active group,

$$\mathbf{g}^\star = \arg \max_{\mathbf{g} \in \mathcal{G}} \pi_{\mathbf{g}}^{\text{grp}}(\boldsymbol{\xi}), \quad (36)$$

and separate channel routers then select the two highest-scoring routed experts for velocity and pressure. In each active group and channel, one shared expert is always evaluated and six routed experts are available. If $\mathcal{T}_2^c$ is the selected set for channel $c \in \{u, p\}$, the channel output is

$$\boldsymbol{\rho}_{\theta_r}^c(\boldsymbol{\xi}) = \omega_0^c \mathcal{E}_{0,\mathbf{g}^\star}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}_2^c} \omega_e^c \mathcal{E}_{e,\mathbf{g}^\star}^c(\boldsymbol{\xi}), \quad \omega_0^c + \sum_{e \in \mathcal{T}_2^c} \omega_e^c = 1. \quad (37)$$

Thus one group, one shared expert, and two routed experts per output channel are active for one evaluation. Velocity and pressure routing remain separate because the two channels correct different physical operators.

Each physics-aware expert has a deep nonlinear branch, a linear branch, and a rank-four low-rank quadratic branch:

$$\mathcal{E}(\boldsymbol{\xi}) = \mathcal{N}_{\text{deep}}(\boldsymbol{\xi}) + W_{\text{lin}}\boldsymbol{\xi} + \kappa W_o \left[(U^\top \boldsymbol{\xi}) \odot (V^\top \boldsymbol{\xi}) \right], \quad \text{rank}(U) = \text{rank}(V) = 4, \quad \kappa = 0.05. \quad (38)$$

The quadratic branch provides a low-rank correction compatible with the polynomial character of reduced convective dynamics, while the Galerkin operator remains explicitly available through (24).

3.3. Regime-specific dynamical contracts

Steady-local specialist. The steady model retains its indexed phase features, advances velocity using its native explicit-Euler step,

$$\mathbf{a}_{n+1}^{(S)} = \mathbf{a}_n^{(S)} + \Delta t_S \left[\mathbf{f}_S^G + \boldsymbol{\rho}_{\theta_S}^\mu \right]_n, \quad (39)$$

and applies the additive pressure closure (27). Its training objective augments state- and field-level prediction terms with two steady-regime mechanisms. For paired contraction, two trajectories are initialized at the train-derived fixed point $\mathbf{z}^\star$ and at a perturbed state,

$$\mathbf{z}_k^0 = \mathcal{F}_{\theta_S}^{(k)}(\mathbf{z}^\star), \quad \mathbf{z}_k^\delta = \mathcal{F}_{\theta_S}^{(k)}(\mathbf{z}^\star + \delta \mathbf{z}), \quad \mathbf{z} = [\mathbf{a}, \mathbf{b}]. \quad (40)$$

With local coefficient scales σ_S^a and σ_S^b , define

$$d_0^2 = \text{mean} \left[(\delta \mathbf{a} \oslash \sigma_S^a)^2 \right] + \text{mean} \left[(\delta \mathbf{b} \oslash \sigma_S^b)^2 \right] + \varepsilon, \quad (41)$$

$$g_k = \frac{1}{d_0} \left\{ \text{mean} \left[((\mathbf{a}_k^\delta - \mathbf{a}_k^0) \oslash \sigma_S^a)^2 \right] + \text{mean} \left[((\mathbf{b}_k^\delta - \mathbf{b}_k^0) \oslash \sigma_S^b)^2 \right] + \varepsilon \right\}^{1/2}. \quad (42)$$

The loss penalizes gains above a prescribed, horizon-dependent envelope:

$$\mathcal{L}_{\text{pair}} = \text{mean}_{k,n} [\max(0, g_k - \rho_k)]^2. \quad (43)$$

A fixed-point anchoring term evaluates one complete deployed step at the same target,

$$\mathcal{L}_{\text{fp}} = \text{mean} \frac{\|\mathcal{F}_{\theta_S}^a(\mathbf{z}^\star) - \mathbf{a}^\star\|_2^2}{d_u} + \text{mean} \frac{\|\mathcal{F}_{\theta_S}^b(\mathbf{z}^\star) - \mathbf{b}^\star\|_2^2}{d_p}. \quad (44)$$

These are local training constraints. They do not establish global contractivity of the deployed map. In the certified training sequence, the paired-contraction stage (S3) and the fixed-point anchoring stage (S4) share the same HPRS-MoE backbone; they are successive regularization stages rather than separately routed deployment models.

Hopf-local specialist. The Hopf model removes indexed phase inputs and describes oscillatory onset through Reynolds-conditioned fluctuation coordinates. Conditional train-derived means and scales define

$$\mathbf{a}'_H = (\mathbf{a}_H - \boldsymbol{\mu}_H^a(\mu)) \oslash s_H^a(\mu), \quad \mathbf{b}'_H = (\mathbf{b}_H - \boldsymbol{\mu}_H^b(\mu)) \oslash s_H^b(\mu). \quad (45)$$

Let $V_H \in \mathbb{R}^{2 \times d_u}$ be the train-derived critical-plane projection. The associated coordinates and normalized radius are

$$\zeta_H = V_H(\mathbf{a}_H - \boldsymbol{\mu}_H^a(\mu)), \quad \varrho_H = \frac{\|\zeta_H\|_2}{s_\varrho(\mu) + \varepsilon}. \quad (46)$$

The critical-plane components, normalized radius, and associated fluctuation descriptors form the 67-dimensional augmentation of the common backbone. Velocity is integrated by native RK4. With $F_H(\mathbf{a}, \mathbf{b}, \mathcal{H}; \mu)$ denoting (23) under the Hopf feature contract, pressure, history, and Reynolds number remain fixed within one macro-step. The four stages are

$$\begin{aligned} \mathbf{k}_1 &= F_H(\mathbf{a}_n, \mathbf{b}_n, \mathcal{H}_n; \mu), \\ \mathbf{k}_2 &= F_H(\mathbf{a}_n + \tfrac{1}{2}\Delta t_H \mathbf{k}_1, \mathbf{b}_n, \mathcal{H}_n; \mu), \\ \mathbf{k}_3 &= F_H(\mathbf{a}_n + \tfrac{1}{2}\Delta t_H \mathbf{k}_2, \mathbf{b}_n, \mathcal{H}_n; \mu), \\ \mathbf{k}_4 &= F_H(\mathbf{a}_n + \Delta t_H \mathbf{k}_3, \mathbf{b}_n, \mathcal{H}_n; \mu), \\ \mathbf{a}_{n+1} &= \mathbf{a}_n + \frac{\Delta t_H}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned} \quad (47)$$

Pressure follows (28). A normal-form radial term regularizes training by comparing the predicted radial evolution with a train-derived onset trend, but the normal-form head is not part of deployed inference.

Periodic-local specialist. The periodic model uses its indexed/native phase contract, native RK4 for velocity, and the adaptive algebraic pressure closure. During rollout, its predicted state and history are fed back while query times and their prescribed native phase indices remain external inputs. Training follows a progressive autonomous-rollout curriculum

$$K \in \{4, 8, 12, 16\}, \quad (48)$$

Table 1: Certified dynamical contracts of the three local specialists.

Specialist	Phase contract	Velocity integrator	Pressure closure	Regime-specific mechanism
Steady	indexed	explicit Euler	additive	paired contraction; fixed point
Hopf	phase-free	RK4	adaptive gate	critical-plane radial regularization
Periodic	indexed/native	RK4	adaptive gate	autonomous-rollout curriculum

so that the same learned residual is exposed to increasing closed-loop horizons. The certified periodic model does not use an attractor-conditioned adapter, a radius label, a prototype classifier, attractor cross-entropy, or explicit periodic-class supervision.

3.4. Independent training and model freezing

The three specialists are trained independently in their own charts. A generic local objective is

$$\mathcal{L}_r = \lambda_a \mathcal{L}_a + \lambda_b \mathcal{L}_b + \lambda_u \mathcal{L}_u + \lambda_p \mathcal{L}_p + \lambda_{\text{roll}} \mathcal{L}_{\text{roll}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}} + \mathcal{L}_r^{\text{regime}}, \quad (49)$$

where the first four terms measure modal and reconstructed-field errors, $\mathcal{L}_{\text{roll}}$ is evaluated under autonomous native rollout, $\mathcal{L}_{\text{route}}$ regularizes the inner sparse routers, and $\mathcal{L}_r^{\text{regime}}$ contains only the mechanisms described in Section 3.3. Validation checkpoint selection uses complete held-out Reynolds-number trajectories. The test population does not determine loss weights, training duration, thresholds, or checkpoint selection.

After validation selection, the checkpoint and all associated local assets are frozen. Subsequent training of the outer regime router or the S – H fusion gate does not update the POD bases, means, scalars, Galerkin tensors, Pressure–Poisson operators, inner routers, experts, integrators, or pressure closures. This freezing policy makes the hierarchical system an assembly of independently certified local dynamical models. Exact layer widths, all loss weights, optimizer settings, checkpoint indices, asset hashes, and hardware details are reproducibility information and are reserved for an appendix.

4. Admissibility-Constrained Hierarchical RAL-MoE-ROM

The frozen specialists are assembled by an outer sparse mechanism that separates three decisions: regime preference, topological legality, and dynamical feasibility. Router probabilities alone do not authorize a specialist to participate in a prediction. The resulting hierarchy uses Top-1 routing in regime cores, permits an output-level Top-2 mechanism only on a certified adjacent overlap, and fails closed when no candidate satisfies its native contract.

4.1. Trajectory-level routing and adjacency topology

The outer E2 router receives only the standardized Reynolds number,

$$\tilde{\mu} = \frac{\mu - \mu_R}{\sigma_R}, \quad \pi(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu})) = [\pi_S, \pi_H, \pi_P]^\top. \quad (50)$$

One routing decision is made for a complete fixed- μ trajectory. The outer router is therefore a trajectory-level, temporal-consistent regime router, rather than a per-step switching policy. Because the available training data do not constitute certified continuous startup or parameter-ramp transitions, the router is not interpreted as a learned physical $S \rightarrow H \rightarrow P$ transition model.

The only legal regime adjacency is

$$A_{\text{adj}} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad (51)$$

using the class order (S, H, P) . For an anchor regime k , the topological candidate set is

$$C_{\text{top}}(k) = \{j \in \mathcal{R} : A_{\text{adj},kj} = 1\}. \quad (52)$$

This removes a direct S – P pair and excludes Top-3 activation before any field prediction is evaluated.

4.2. Dynamical admissibility certification

For specialist j , initial physical history $\mathcal{H}$, Reynolds number μ , and registered rollout horizon K , define the binary admissibility indicator

$$m_j(\mathcal{H}, \mu, K) \in \{0, 1\}. \quad (53)$$

The value $m_j = 1$ requires all of the following:

1. every predicted modal state and reconstructed field is finite;
2. the registered horizon contains no numerically divergent window;
3. the local POD, scaler, history, phase, integrator, Pressure–Poisson, and pressure-gauge contracts are satisfied;
4. independent validation trajectories support stable native rollout over the same development domain and horizon; and
5. inference uses neither test truth, a future endpoint, a future state, nor an uncertified cross-chart input.

Given router weights π_j , topology and admissibility produce

$$\tilde{\pi}_j = \frac{\mathbf{1}_{\{j \in C_{\text{top}}\}} m_j \pi_j}{\sum_{k \in \mathcal{R}} \mathbf{1}_{\{k \in C_{\text{top}}\}} m_k \pi_k + \varepsilon}. \quad (54)$$

If two adjacent candidates are admissible, a legal Top-2 comparison may be formed. If only one is admissible, the hierarchy degenerates deterministically to that Top-1 specialist. If none is admissible, the system returns a fail-closed status instead of forcing a numerical prediction. Admissibility is thus a prerequisite for capability comparison, not a loss term that can be compensated by router confidence.

4.3. Boundary-specific hierarchical assembly

Regime-core prediction. In a certified core domain, only its native specialist is executed:

$$\widehat{\mathbf{x}} = \begin{cases} \widehat{\mathbf{x}}_S, & \text{steady core,} \\ \widehat{\mathbf{x}}_H, & \text{Hopf core,} \\ \widehat{\mathbf{x}}_P, & \text{periodic core.} \end{cases} \quad (55)$$

Each prediction is generated in its local chart and reconstructed with its own decoder.

One-sided native-supported S - H fusion. The certified S - H protocol uses only S -native query trajectories. A common three-state physical history $\mathcal{H}_n^{\text{phys}}$ initializes two independent rollouts. The steady branch uses its native indexed phase, explicit-Euler velocity step, and additive pressure closure. The same physical history is projected into the phase-free Hopf chart according to Section 2.3; the Hopf branch then uses its native RK4 velocity step and adaptive pressure closure. Neither rollout receives a future true state.

For each registered $K = 56$ query window, a pair-specific gate is evaluated once from the Reynolds number and eight chart-independent descriptors of the initial physical history. For the three observed states $i \in \{0, 1, 2\}$, define

$$\begin{aligned} E_i^u &= \|\mathbf{u}_i\|_{M_u}^2, & E_i^p &= \|p_i^\circ\|_{M_p}^2, \\ D_i^u &= \frac{\|\mathbf{u}_i - \mathbf{u}_{i-1}\|_{M_u}}{\Delta t_i}, & D_2^p &= \frac{\|p_2^\circ - p_1^\circ\|_{M_p}}{\Delta t_2}, \end{aligned} \quad (56)$$

$$g_i = \frac{\log E_i^u - \log E_{i-1}^u}{\Delta t_i}, \quad \kappa_u = \frac{\left\| \frac{\mathbf{u}_2 - \mathbf{u}_1}{\Delta t_2} - \frac{\mathbf{u}_1 - \mathbf{u}_0}{\Delta t_1} \right\|_{M_u}}{\sqrt{E_2^u} + \varepsilon}. \quad (57)$$

With $|\Omega|$ denoting the weighted domain area, the descriptor vector is

$$\mathbf{d}_n = [d_1, \dots, d_8]^\top, \quad (58)$$

where

$$\begin{aligned} d_1 &= \tfrac{1}{2} \log(E_2^u/|\Omega|), & d_2 &= \tfrac{1}{2} \log(E_2^p/|\Omega|), \\ d_3 &= \log\left(1 + \frac{D_2^u}{\sqrt{E_2^u} + \varepsilon}\right), & d_4 &= \log\left(1 + \frac{D_1^u}{\sqrt{E_1^u} + \varepsilon}\right), \\ d_5 &= \tanh g_2, & d_6 &= \tanh(g_2 - g_1), \\ d_7 &= \log\left(1 + \frac{D_2^p}{\sqrt{E_2^p} + \varepsilon}\right), & d_8 &= \log(1 + \kappa_u). \end{aligned} \quad (59)$$

All norms in (56)–(59) use the common mesh, area weights, and pressure gauge, so the descriptors do not depend on which local chart represents the initial state.

The E2 probabilities are first renormalized on the legal pair,

$$\bar{\pi}_S = \frac{\pi_S^{E2}}{\pi_S^{E2} + \pi_H^{E2}}, \quad \bar{\pi}_H = 1 - \bar{\pi}_S, \quad \ell_{SH}^{E2} = \log \frac{\bar{\pi}_S + \varepsilon}{\bar{\pi}_H + \varepsilon}. \quad (60)$$

Let $\widehat{\mathbf{q}}$ be the standardized version of $[\mu; \mathbf{d}_n] \in \mathbb{R}^9$. A lightweight correction $c_\theta(\widehat{\mathbf{q}})$, whose final layer is initialized to zero, gives

$$\alpha_S = \text{sigmoid}(\ell_{SH}^{E2} + c_\theta(\widehat{\mathbf{q}})), \quad \alpha_H = 1 - \alpha_S. \quad (61)$$

Zero initialization exactly reproduces the frozen E2 pair probability at the start of gate training. The weights satisfy

$$\alpha_S + \alpha_H = 1, \quad \alpha_S \geq 0, \quad \alpha_H \geq 0, \quad (62)$$

and remain constant throughout that window.

The two specialists advance independently to the same physical query times and are decoded before

fusion. The velocity and gauge-corrected pressure outputs are

$$\widehat{\mathbf{u}}_{SH}(t_q) = \alpha_S \widehat{\mathbf{u}}_S(t_q) + \alpha_H \widehat{\mathbf{u}}_H(t_q), \quad (63)$$

$$\widehat{p}_{SH}^\circ(t_q) = \alpha_S \widehat{p}_S^\circ(t_q) + \alpha_H \widehat{p}_H^\circ(t_q). \quad (64)$$

The gate is trained on the training population and selected on validation data using the same sealed native-output cache. For $c \in \{u, p\}$, let $\mathbf{e}_S^c = \widehat{\mathbf{x}}_S^c - \mathbf{x}_{\text{true}}^c$ and $\mathbf{e}_H^c = \widehat{\mathbf{x}}_H^c - \mathbf{x}_{\text{true}}^c$. The convex field error is

$$Q^c(\alpha_S) = \alpha_S^2 \|\mathbf{e}_S^c\|_{M_c}^2 + (1 - \alpha_S)^2 \|\mathbf{e}_H^c\|_{M_c}^2 + 2\alpha_S(1 - \alpha_S) \langle \mathbf{e}_S^c, \mathbf{e}_H^c \rangle_{M_c}, \quad (65)$$

and the implemented objective is

$$\mathcal{L}_{T2C} = \text{mean}_{w,k} \left[\frac{Q^u(\alpha_S^{(w)})}{\|\mathbf{u}_{\text{true}}\|_{M_u}^2 + \varepsilon} + \frac{Q^p(\alpha_S^{(w)})}{\|p_{\text{true}}^\circ\|_{M_p}^2 + \varepsilon} \right]. \quad (66)$$

Future attractor truth is not an online gate input. Attractor diagnostics are reserved for frozen scientific evaluation and validation-stage interpretation. The fusion is strictly S-native, window-conditioned, and output-only. It is not modal-coefficient fusion, normalized-state fusion, continuous-RHS fusion, per-step gating, or a bidirectional S-native/H-native protocol. The fused output is not fed back into either specialist.

H-P admissibility-driven fallback. In the certified P-native development domain, the periodic specialist supports finite native rollout, whereas the frozen Hopf specialist lacks a validation-supported common feasible band over the registered horizon. Consequently,

$$m_H = 0, \quad m_P = 1, \quad (67)$$

and (54) gives

$$\widetilde{\pi}_H = 0, \quad \widetilde{\pi}_P = 1, \quad \widehat{\mathbf{x}}_{HP} = \widehat{\mathbf{x}}_P. \quad (68)$$

No *H-P* convex gate is trained and no soft-fusion gain is claimed. This P-only/native-time hard route is a formal fail-closed output of the same admissibility principle that permits *S-H* fusion; it is not treated as an incomplete fusion experiment.

Let $\Omega_S, \Omega_{SH}, \Omega_H, \Omega_{HP}, \Omega_P$ denote the five operating regions frozen from training and validation evidence. Combining the preceding cases, the physical-field predictor is

$$\widehat{\mathbf{x}}^{\text{phys}} = \begin{cases} \mathcal{D}_S \circ \mathcal{F}_S, & (\mathcal{H}, \mu) \in \Omega_S, \\ \alpha_S \mathcal{D}_S \circ \mathcal{F}_S + \alpha_H \mathcal{D}_H \circ \mathcal{F}_H, & (\mathcal{H}, \mu) \in \Omega_{SH}, \\ \mathcal{D}_H \circ \mathcal{F}_H, & (\mathcal{H}, \mu) \in \Omega_H, \\ \mathcal{D}_P \circ \mathcal{F}_P, & (\mathcal{H}, \mu) \in \Omega_{HP} \cup \Omega_P, \\ \text{FAIL-CLOSED}, & \sum_{j \in \mathcal{C}_{\text{top}}} m_j = 0. \end{cases} \quad (69)$$

In the second branch, the sum is taken only after both local trajectories have been decoded into velocity and gauge-corrected pressure fields; it does not define a sum of local modal states.

4.4. Offline-online workflow and certified methodological scope

Algorithm 1: Offline construction of RAL-MoE-ROM

Input: Complete Reynolds-number trajectories and registered native specialist contracts.
Output: Frozen local specialists, outer router, admissibility records, and the certified $S-H$ gate.

Partition complete trajectories into disjoint training, validation, and test Reynolds-number populations;
Construct train-only local velocity and pressure POD charts, means, scalars, Galerkin operators, and Pressure-Poisson assets;
Train the S , H , and P HPRS-MoE-ROM specialists independently;
Select each specialist with validation trajectories only and freeze all local assets;
Train and freeze the E2 trajectory-level outer router;
Apply the adjacency topology in (51);
Certify each candidate specialist on its registered development domain and horizon;
Train T2-C only on the validation-supported, S-native $S-H$ overlap;
Freeze all thresholds, gates, and hierarchical decisions before final evaluation;

Algorithm 2: Hierarchical online prediction with RAL-MoE-ROM

Input: μ , a valid three-state physical history, query times, and a registered horizon K .
Output: Physical velocity, gauge-corrected pressure, active specialists, weights, and admissibility diagnostics.

Evaluate the trajectory-level E2 router once;
restrict the candidates by the adjacency topology;
evaluate the admissibility mask for the candidate specialists;
if no candidate is admissible then
 | return fail-closed diagnostics without a forced prediction;
else
 | if exactly one candidate is admissible then
 | execute its frozen native rollout and decode the physical fields;
 | else
 | verify that the pair is the certified S-native $S-H$ pair;
 | map the common physical history into both local charts;
 | execute two independent frozen native rollouts at aligned query times;
 | evaluate the T2-C gate once and keep its weights fixed over the window;
 | decode both predictions and fuse only the physical outputs by (63)–(64);
return predictions and the complete routing/admissibility record;

Under Top-1 routing, one frozen specialist is active. The legal $S-H$ Top-2 case evaluates two frozen native rollouts; the gate cost is small relative to this second rollout. No common POD decoder is needed and no specialist parameter is updated online. Exact wall-clock acceleration factors are not inferred without direct timing measurements.

The certified methodological scope can now be stated precisely. RAL-MoE-ROM does not introduce a common public POD state, direct cross-chart modal blending, a full-state continuous vector field, a pressure derivative, cross-specialist RHS fusion, fused-state feedback, a direct $S-P$ route, Top-3 activation, or per-step outer regime switching. Steady uses native explicit Euler, while Hopf and periodic velocity use native RK4; pressure remains a macro-step algebraic closure. Steady and periodic phase inputs are indexed/native, whereas Hopf is phase-free. The only soft combination is the S-native, window-conditioned, output-only $S-H$ field fusion, and the $H-P$ boundary degenerates to P-only under admissibility. These restrictions define the verified model, rather than engineering omissions to be silently replaced by stronger untested claims.

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